Printed Pages:02 **Sub Code:KCA-034**

Paper Id: 236506 **Roll No.** | | | | | | | | | | |

MCA
(SEM IV) THEORY EXAMINATION 2022-23
DATA ANALYTICS

Time: 3 Hours ***Total Marks: 100***

Note: Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt*all* questions in brief. **2 x 10 = 20**

a. Explain various applications of data analytics.

b. Discuss competitive learning.

c. Explain stream computing.

d. Explain CLIQUE.

e. Name any four NoSQL databases.

f. Explain R tool graphical user interface.

g. Explain the method of mining frequent itemset.

h. Explain the reason why Hadoop comes in market.

i. Explain stochastic search method.

j. What are key roles for successful analytic projects.

SECTION B

2. Attempt any *three* of the following: **10x3=30**

a. Explain various phases of data analytics life cycle.

b. Explain rule induction with suitable example.

c. Explain real time sentiment analysis.

d. Discuss k-means clustering technique with suitable example.

e. Explain HDFS architecture.

SECTION C

3. Attempt any *one* part of the following: **10x1=10**

a. Explain data, characteristics of data, and types of data.

b. Explain the concept of Principal Component Analysis (PCM).

QP23EP1_311 | 17-06-2023 14:05:25 | 115.243.172.58

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4. Attempt any *one* part of the following: 10x1=10

a. Discuss Real -time analytics platform. Also, explain stock market prediction case study.

b. Explain Apriori algorithm with example.

5. Attempt any *one* part of the following: 10x1=10

a. Discuss NoSQL database and its types.

b. Explain hierarchical clustering technique with example.

6. Attempt any *one* part of the following: 10x1=10

a. Differentiate between
   (1) HDFS and HBase
   (2) HBase and HIVE

b. Explain clustering in non-euclidean space.

7. Attempt any *one* part of the following: 10x1=10

a. Differentiate between
   (1) MapReduce and YARN
   (2) Linear regression and Logistic regression

b. Explain bayes theorem and concept learning.

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